

NOLAN KOCH

Pittsburgh, PA (Open to Remote) | 412-215-1748 | kochnolan376@gmail.com | [GitHub](#) | [Portfolio](#) | English & Arabic

EDUCATION

Slippery Rock University	— B.S. Cybersecurity (Secure Software Development), Expected Dec 2026	<i>Relevant Coursework:</i> Data Structures & Algorithms, Networking, Software Assurance, Software Engineering
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TECHNICAL SKILLS

Languages & Frameworks: Python, C++, Java, JavaScript/TypeScript, SQL, Bash, React, PyTorch, LangChain	Security & Tools: Linux (RHEL/Ubuntu), Docker, Burp Suite, Kali Linux, NIST/FedRAMP, Git, CI/CD, AWS/GCP	Core Competencies: Vulnerability Management, RBAC/IAM, Threat Modeling, Incident Response, OSINT
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EXPERIENCE

Qintel |

Full-Stack Software Engineering Intern

May 2026 – Present

- Engineered responsive data visualizations using **React** and **Node.js** APIs, empowering **100,000+** active users to analyze complex cyber intelligence datasets.
- Implemented a query scope guard that cancels searches exceeding a result threshold and prompts the user to refine their query.
- Shipped **full-stack features end-to-end** across the SDLC, resolving **UI/UX bugs** to ensure ease of use for enterprise-grade products utilized daily by global customers.

Broma |

Full-Stack Engineer

Oct 2025 – May 2026

- Architected a multi-tenant **B2B SaaS** portal using **React** and **Supabase**, delivering a custom **CRM** dashboard for vendors to manage event campaigns, analytics, and digital presence.
- Engineered a **semantic search** pipeline with **Google Gemini** and vector embeddings, enabling **natural language discovery** (e.g., "quiet study spots") to replace rigid keyword matching.
- API Defense:** Developed edge functions with **Zod validation** and rate-limiting to mitigate **SQL Injection** and **DoS attacks** against AI endpoints.
- Designed a proprietary feed-ranking algorithm in **PostgreSQL** to prioritize local businesses, directly translating the go-to-market strategy into software logic for release.
- Published** app to IOS app store [See here](#)

Westinghouse Electric Company |

Linux Systems Administrator Intern

May 2025 – May 2026

- Automated Remediation:** Scripted Python/Bash workflows to patch CVEs across **50+ RHEL nodes**, reducing zero-day exposure time by **30%**.
- Compliance Enforcement:** Collaborated with cross-functional security teams to audit system configurations against **NIST 800-53** baselines, ensuring infrastructure audit-readiness.
- SIEM Optimization:** Configured log aggregation pipelines to enhance visibility for incident response and proactive threat hunting.

LEADERSHIP & PROJECTS

Trusted CI Scholars Program |

Scholar

May 2026 – Nov 2026

- Selected for a competitive National Science Foundation (NSF) cybersecurity fellowship program.
- Representing the cohort at the NSF Cybersecurity Summit in Irvine, California. [Trusted CI](#)

Slippery Rock University Cybersecurity Club |

Vice President

- Team Leadership:** Lead participation in **Capture The Flag (CTF)** events (Web Exploitation, Reverse Engineering, Cryptography).
- Education & Mentorship:** Design and teach bi-weekly workshops on **OWASP Top 10** and ethical hacking